

# Wearables and Habits: Setup, Data Download and Package Audit

Issue #3

3 October 2026

## 1 Summary

The existing research container builds and runs. Open e-commerce 1.0 (CC0) is downloaded and checksum-verified. The dataset matches the plan's description: Amazon purchases linked to a survey by respondent ID. The staggered-DiD stack (R `did`, `fixest`, `HonestDiD`, `didimputation`, `bacondecomp`; Python `pyfixest`) is added to the same container. No new container was needed. This is a setup report; the full audit and go/no-go are the next issue.

## 2 Data

Source: Harvard Dataverse, DOI [10.7910/DVN/YGLYDY](https://doi.org/10.7910/DVN/YGLYDY), license CC0 1.0. Downloaded by `code/scripts/download_open_ecommerce.py` (retry/backoff, HTTP range resume, md5 check) to `code/build/input/public/open_ecommerce/` (gitignored). All md5 sums match Dataverse.

| File                                                                                      | Size   | Content                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>amazon-purchases.csv</code>                                                         | 313 MB | Order date, unit price, quantity, ship-to state, title, ASIN, category, respondent ID                                                                                                      |
| <code>survey.csv</code>                                                                   | 1.3 MB | One row per respondent: age group, race, education, income, gender, state, household size, order frequency, smoking/alcohol/diabetes/wheelchair, 2021 life changes, data-sharing attitudes |
| <code>fields.csv</code>                                                                   | 2.5 kB | Question text for survey columns                                                                                                                                                           |
| <code>survey-instrument.pdf</code><br>and<br><code>prescreen-survey-instrument.pdf</code> | 1.7 MB | Questionnaires                                                                                                                                                                             |

Download quirk: `survey.csv` is ingested by Dataverse as a tab file; the listed md5 refers to the *original* upload, so the script requests `?format=original`. Dataverse also returns 403 to requests without a User-Agent.

### First look (crude keyword flags, not the audit classification)

Wearable = title contains Fitbit, Garmin, Apple Watch, smartwatch, fitness/activity tracker or Forerunner. Running = title/category keywords. The wearable flag includes accessories (watch bands are 43% of flagged rows), so it overstates device purchases. Script: `code/scripts/first-look_open_ecommerce.py`.

| Item                                         | Value                    |
|----------------------------------------------|--------------------------|
| Purchase rows                                | 1,850,717                |
| Users with purchases                         | 5,027                    |
| Survey respondents                           | 5,027                    |
| Order dates                                  | 2018-01-01 to 2024-08-15 |
| Median purchases per user                    | 232                      |
| Rows flagged wearable (title keywords)       | 6,735                    |
| Users with a flagged wearable (ever-treated) | 1,982                    |
| of which $\geq 12$ months pre-history        | 1,413                    |
| of which $\geq 12$ months pre and post       | 1,076                    |
| Rows flagged running-related                 | 10,827                   |
| Users with a running-related purchase        | 2,802                    |

  

| First flagged wearable purchase, by year: |      |      |      |      |      |
|-------------------------------------------|------|------|------|------|------|
| 2018                                      | 2019 | 2020 | 2021 | 2022 | 2023 |
| 463                                       | 422  | 359  | 406  | 308  | 24   |

Takeaways for the audit: the data run 2018–Aug 2024; there are roughly 2,000 ever-treated users and about 1,100 with at least a year of history on both sides under the crude flag; cohorts are spread over 2018–2022 (few in 2023). The category field (e.g. `WEARABLE_COMPUTER`, `BIOMETRIC_MONITOR`, `SHOES`) should give a cleaner classification than titles. The survey has no running or exercise items, so the proxy cannot be validated within the dataset.

### 3 Packages

| Need                                      | Package                                                         | Status                    |
|-------------------------------------------|-----------------------------------------------------------------|---------------------------|
| Wrangling large CSV                       | pandas, polars, pyarrow                                         | already in container      |
| Panel/FE, baseline TWFE                   | linearmodels, statsmodels                                       | already in container      |
| Callaway–Sant’Anna (primary)              | R <code>did</code> 2.5.1                                        | added                     |
| Sun–Abraham, fast FE                      | R <code>fixest</code> 0.14.2; Python <code>pyfixest</code> 0.60 | added                     |
| Imputation estimator (robustness)         | R <code>didimputation</code> 0.5.1                              | added                     |
| TWFE weight diagnostics                   | R <code>bacondecomp</code> 0.1.1                                | added                     |
| Sensitivity to parallel-trends violations | R <code>HonestDiD</code> 0.2.8                                  | added (GitHub, see below) |
| Plots                                     | matplotlib, seaborn, R <code>ggplot2</code>                     | already in container      |
| Structural extension (if gated in)        | pyblp                                                           | already in container      |

Choice: R is the reference implementation for `did` and `HonestDiD`, so estimation runs in R inside the same container, with Python for data prep and a `pyfixest` cross-check. Everything was added to `code/Dockerfile`; `requirements.txt` is empty and was left unchanged.

Install pitfalls: `HonestDiD` is not on CRAN (installed from GitHub). Its dependency `Rglpk` only builds against conda’s `glpk` (system `libglpk-dev` is not picked up), and `CVXR`  $\geq 1.1$  needs a Rust toolchain, so `CVXR` is pinned to 1.0-15. `arrow` (R) failed to install and is not needed; use `pyarrow`.

## 4 Open items

- Docker compose warns that `version` is obsolete and that an orphan container from another project exists; harmless, not changed.
- Mount of `analysis/` in the compose file is needed to write outputs from the container; it is already present.
- Next issue: data audit (clean wearable vs. accessory classification, running-gear classification, cohort sizes, pre-trends) and go/no-go.